Git & GitHub Interview Questions and Answers

Q1. What is Git?

Git is a distributed version control system (DVCS) used to track code changes, collaborate with teams, manage branches, and maintain version history.

Q2. What is GitHub?

GitHub is a cloud platform for hosting Git repositories and collaborating using pull requests, issues, actions (CI/CD), and project boards.

Q3. Common Git Commands

git init, git clone, git status, git add, git commit, git log, git branch, git checkout, git merge, git push, git pull, git fetch.

Q4. Difference Between Git and GitHub

Git is a version control tool that runs locally. GitHub is a cloud platform that hosts Git repositories and provides collaboration features.

Q5. What is a Branch in Git?

A branch is a lightweight movable pointer to a commit that allows independent development of features or fixes.

Q6. How Do You Create and Switch to a New Branch?

Use 'git branch ' to create, 'git checkout ' to switch, or 'git checkout -b ' to create and switch in one command.

Q7. What is Merging in Git?

Merging combines changes from one branch into another. Example: git checkout main, then git merge .

Q8. What is a Merge Conflict?

A merge conflict occurs when Git cannot automatically reconcile differences between branches. Resolve conflicts manually, then add and commit the changes.

Q9. What is a Pull Request?

A pull request is a request to review and merge changes from one branch into another.

Q10. What is Git Stash?

Git Stash temporarily saves uncommitted changes so you can work on something else and reapply them later.

Q11. Difference Between git reset, git revert, and git restore

git reset moves HEAD and can rewrite history. git revert creates a new commit that undoes changes. git restore discards changes in the working directory or staging area.

Q12. What is Fork in GitHub?

Forking creates a copy of a repository under your GitHub account, commonly used for contributing to open-source projects.

Q13. What is the Purpose of .gitignore?

.gitignore specifies files and folders that Git should ignore and not track.

Q14. What is a Git Tag?

A tag marks a specific point in repository history, usually for release versions.

Q15. What is a Remote in Git?

A remote is a version of your repository hosted on a network or internet service such as GitHub.

Q16. Difference Between git pull and git fetch

git fetch downloads changes without merging. git pull downloads and automatically merges changes into the current branch.

Q17. How Do You Resolve Merge Conflicts?

Identify conflicting files, edit the conflict markers, keep the correct code, run git add , and commit the merge.

Q18. What is GitHub Actions?

GitHub Actions is a CI/CD service that automates workflows such as testing, building, and deployment.